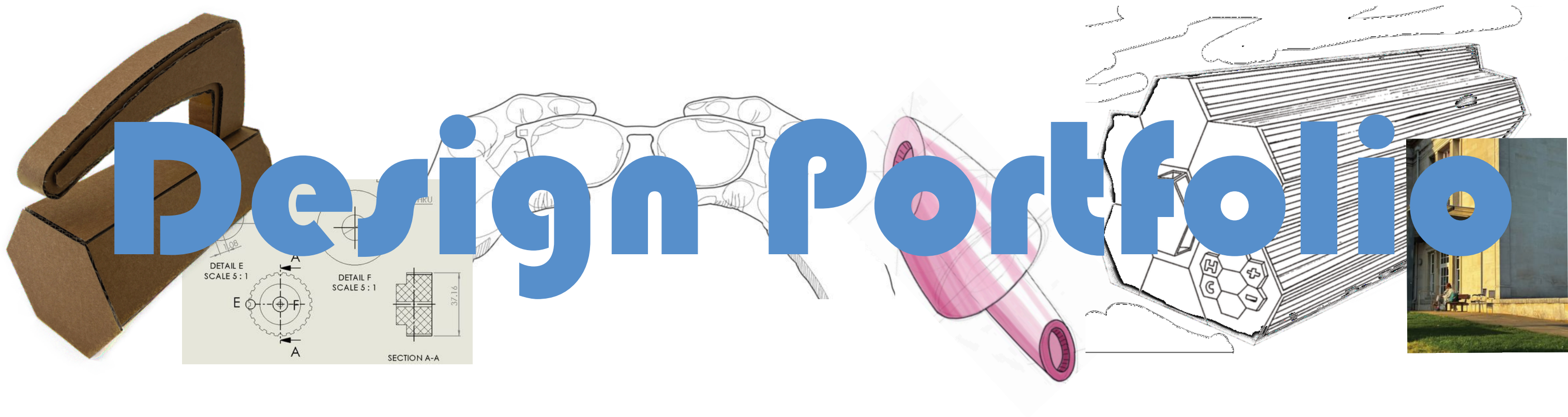




Rachel Zinkin

Drawing

Pages 4-8

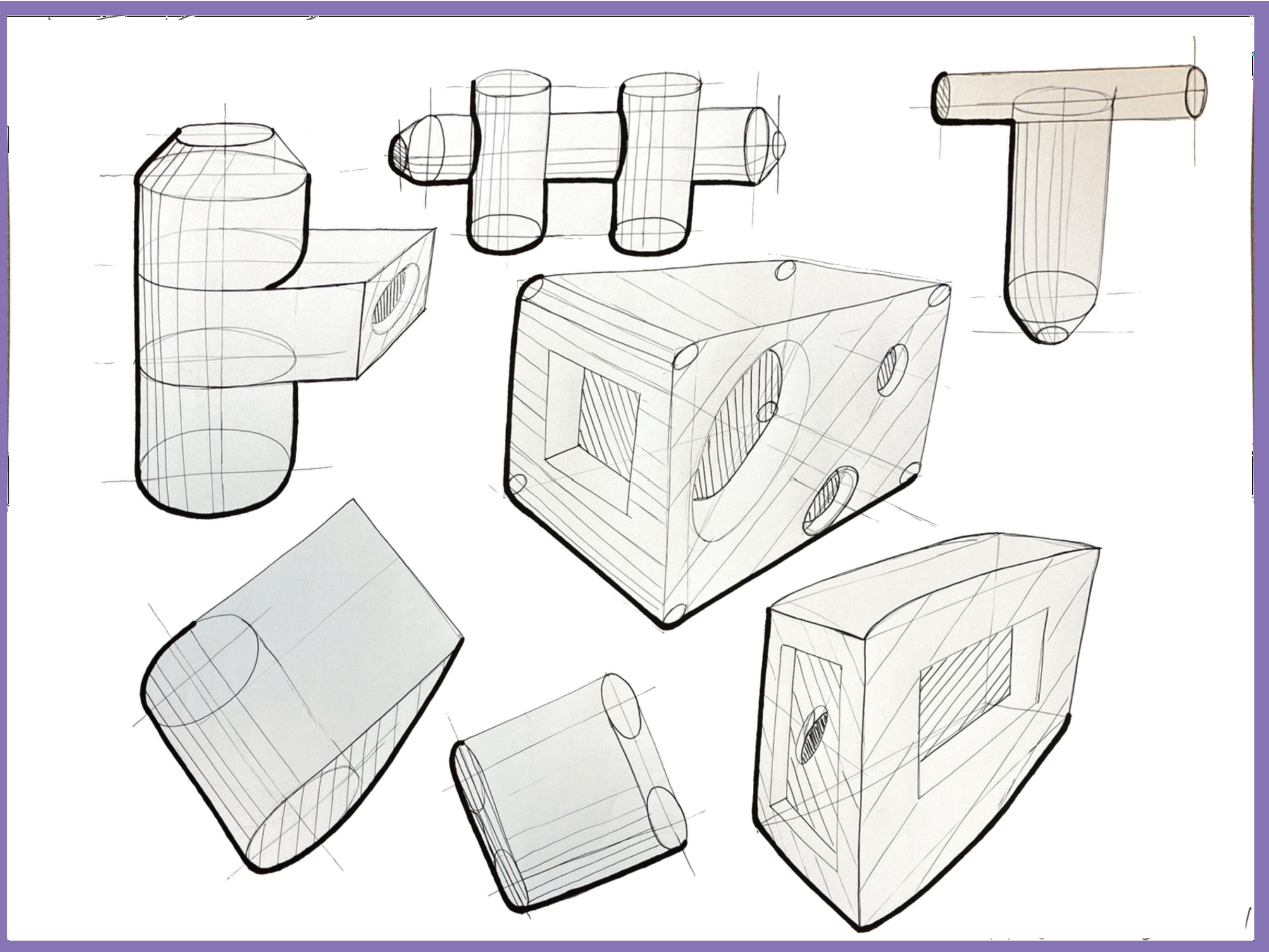
Project work

Pages 10-17

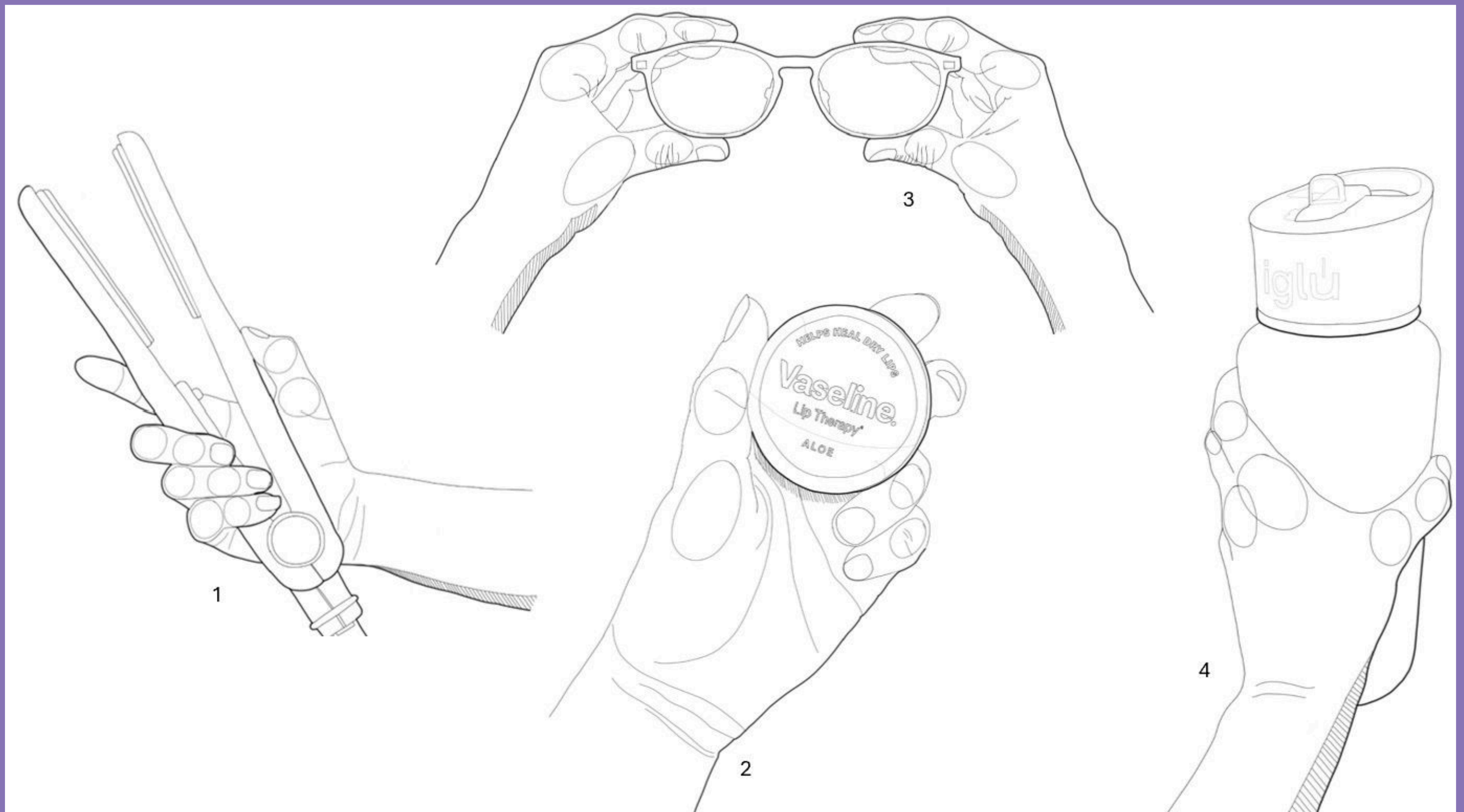
Technical skills

Pages 19-23

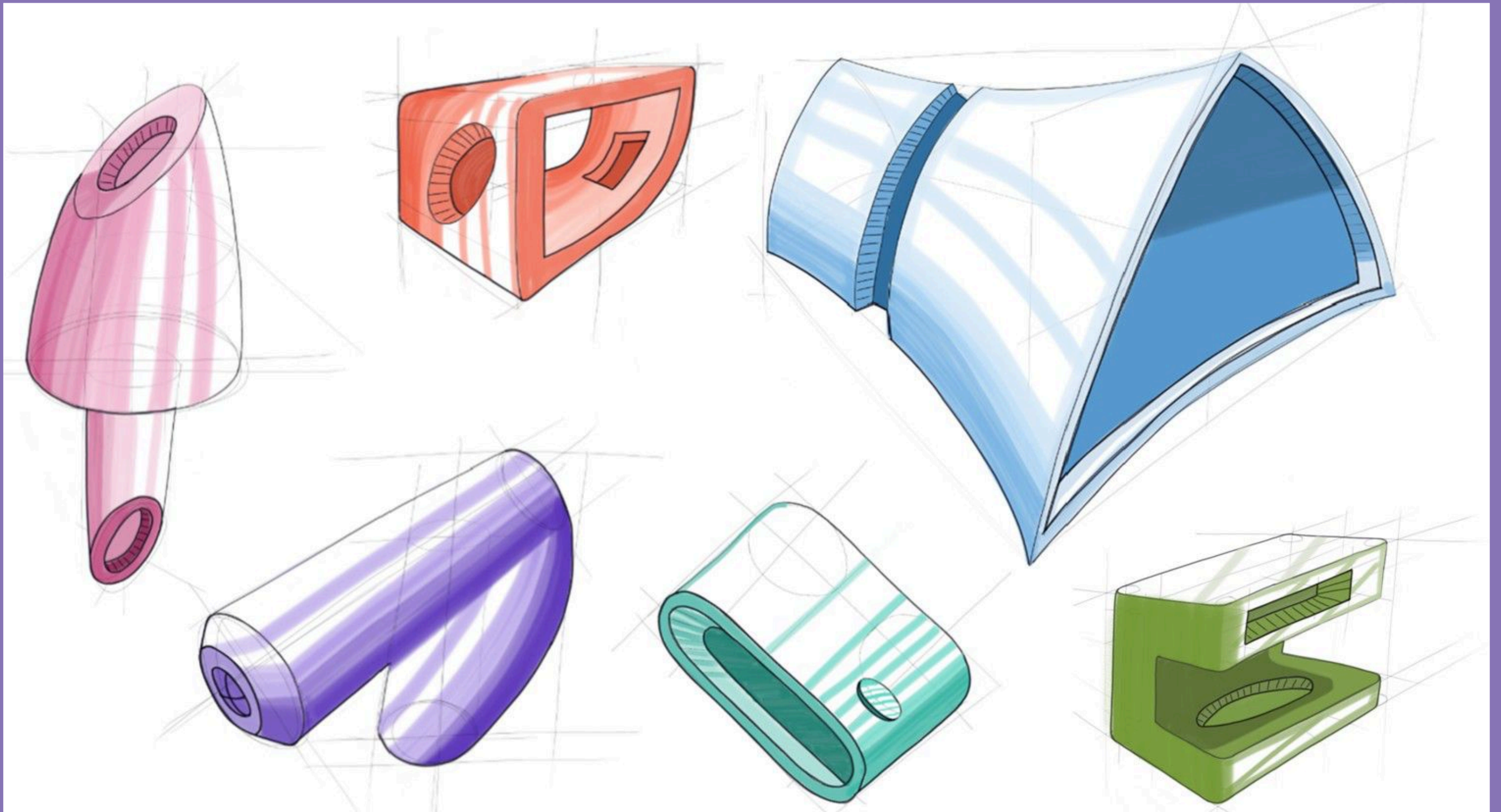
Drawing skills



Level 5 of weekly drawing tests where you must complete the prior level to move on. Here I explored how to draw shapes with circular features and radii in perspective.

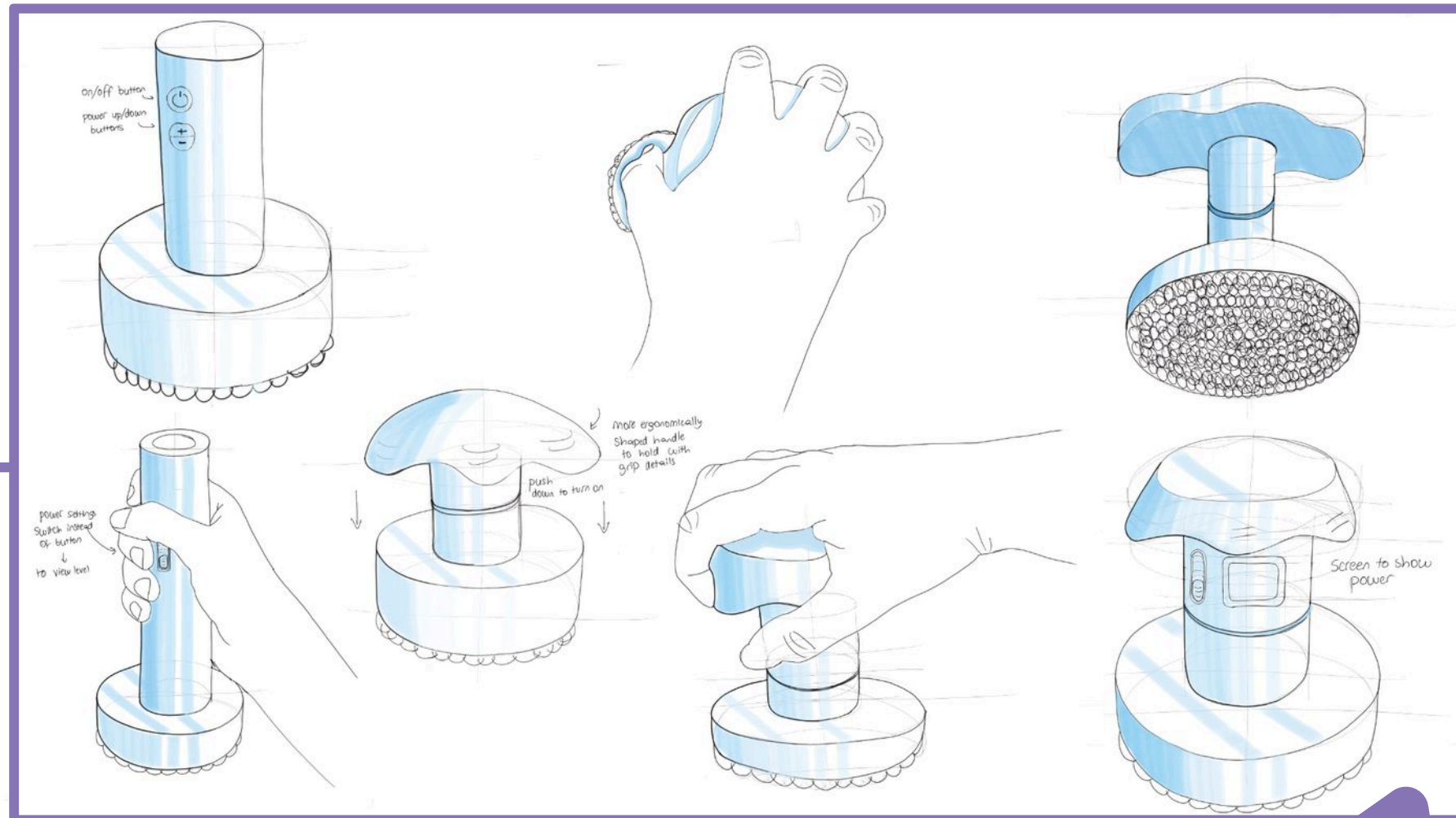
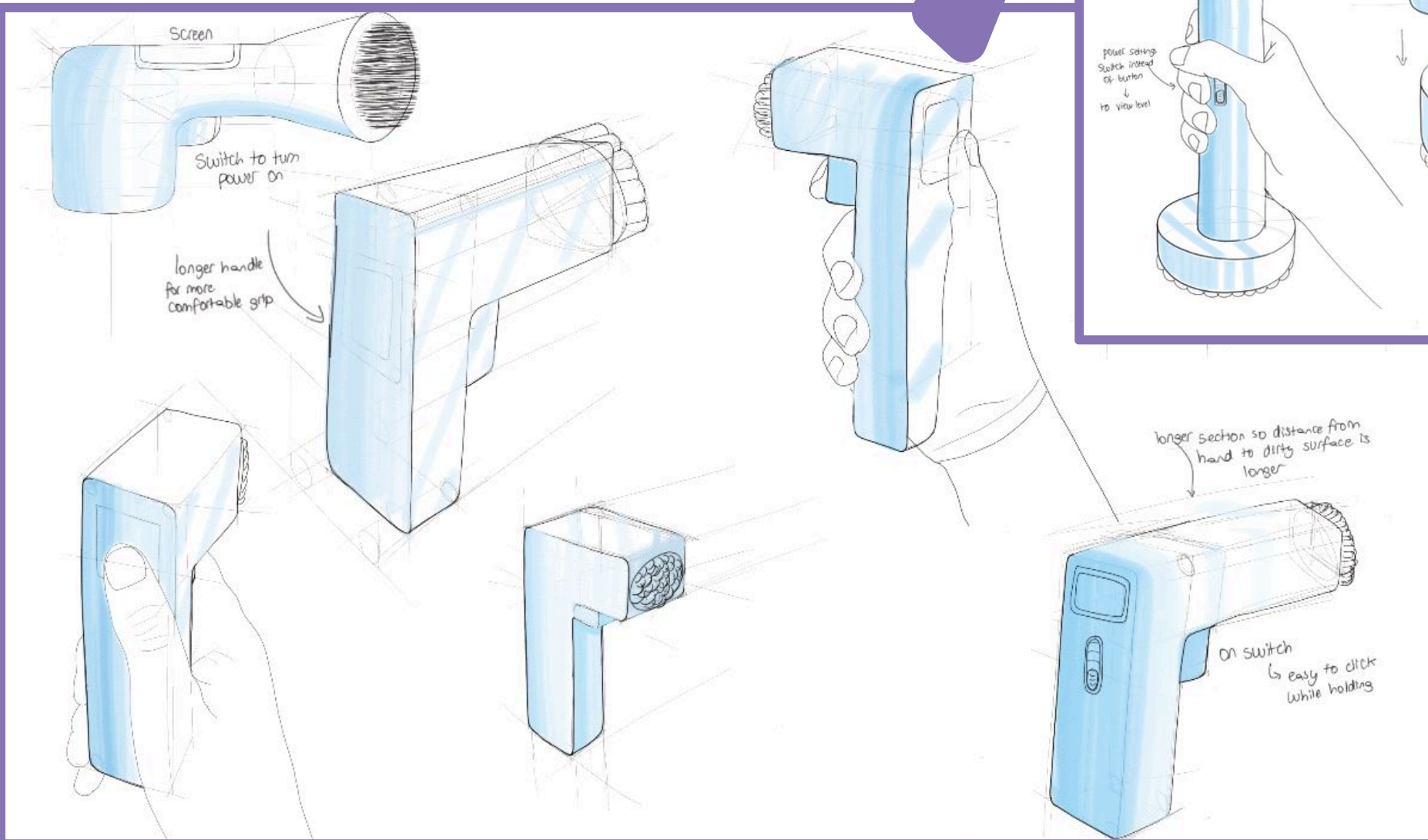


Digital drawing assignment to explore ergonomics of products and human interface.

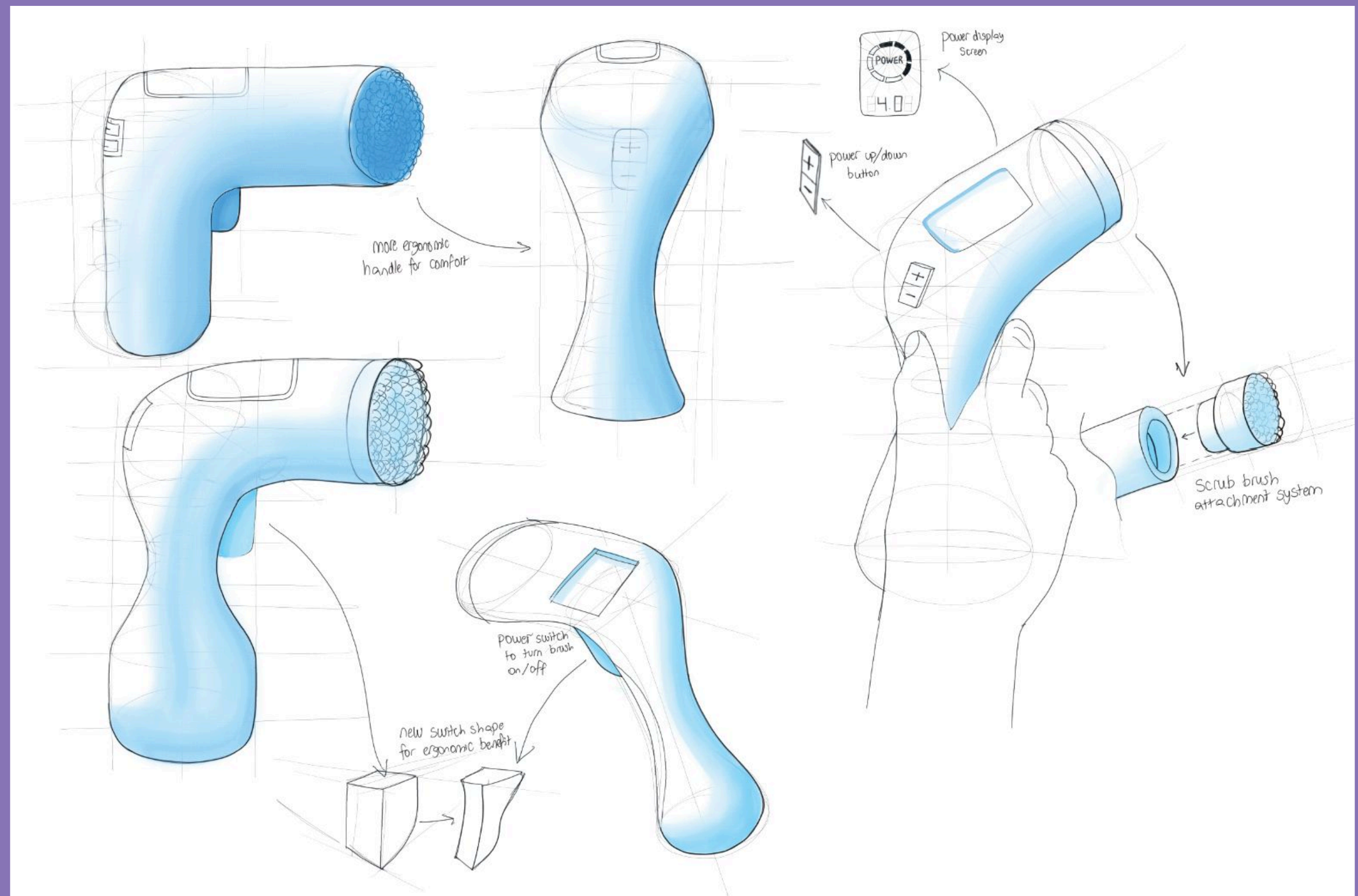


Digital drawing using marker to depict light and shade.

Concept development 1



Concept development 2



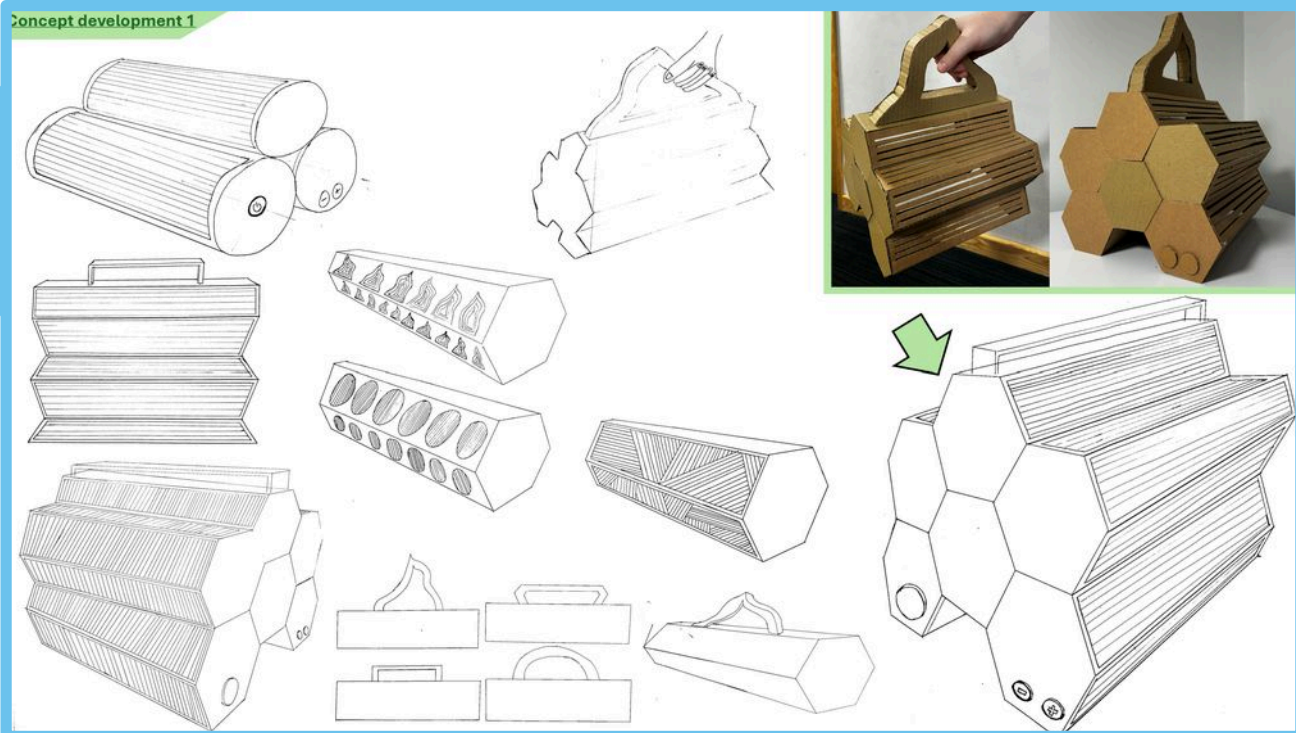
Project work

A hand is holding a cardboard box that has been modified with several horizontal slats. The box is brown and the slats are cut into the top and side panels. The hand is visible on the left side, holding the box. A large, light blue rectangular box with a thin blue border is overlaid on the center of the image, containing the text 'Heater project' in a bold, blue, sans-serif font.

Heater project

Project work: Heater project - ideation + refinement drawing

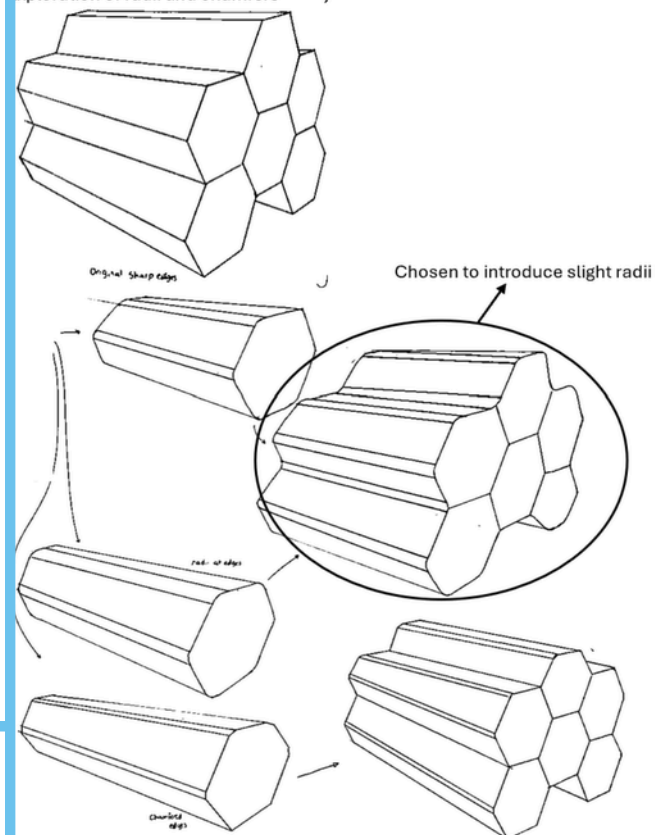
Concept development 1



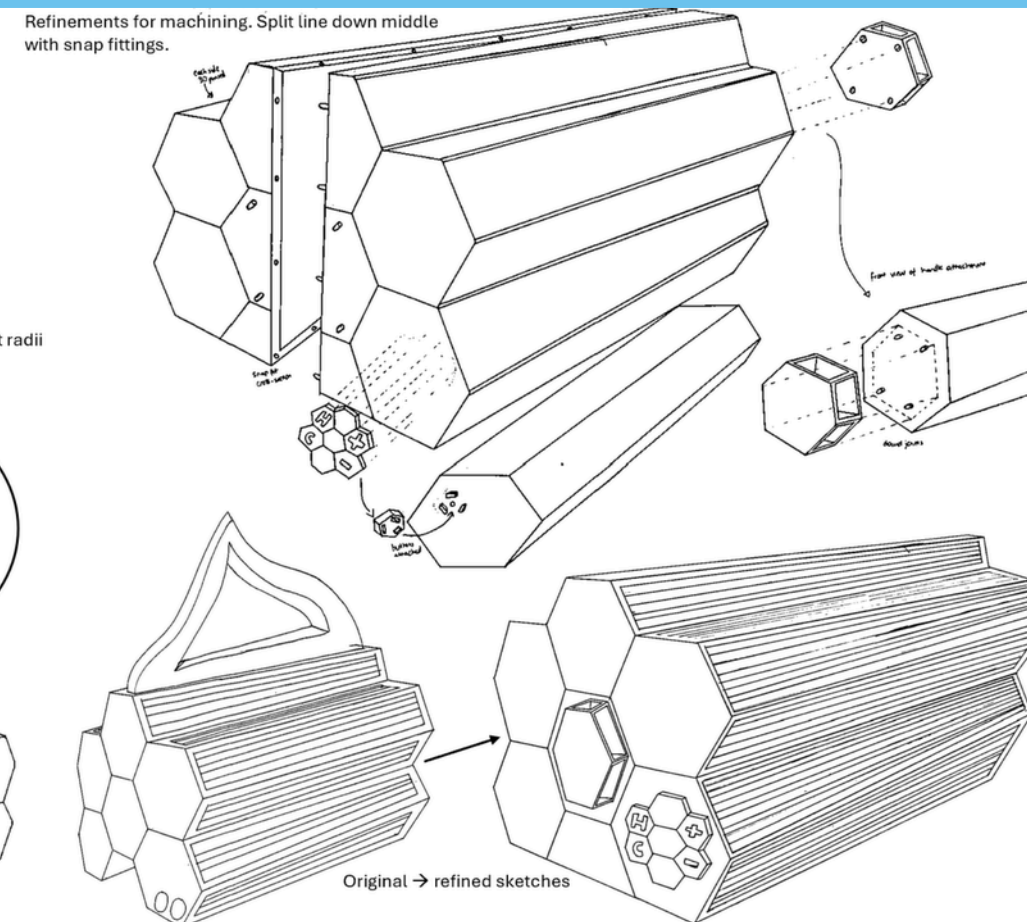
First stage of development of heater idea

Concept refinement 2

Exploration of radii and chamfers

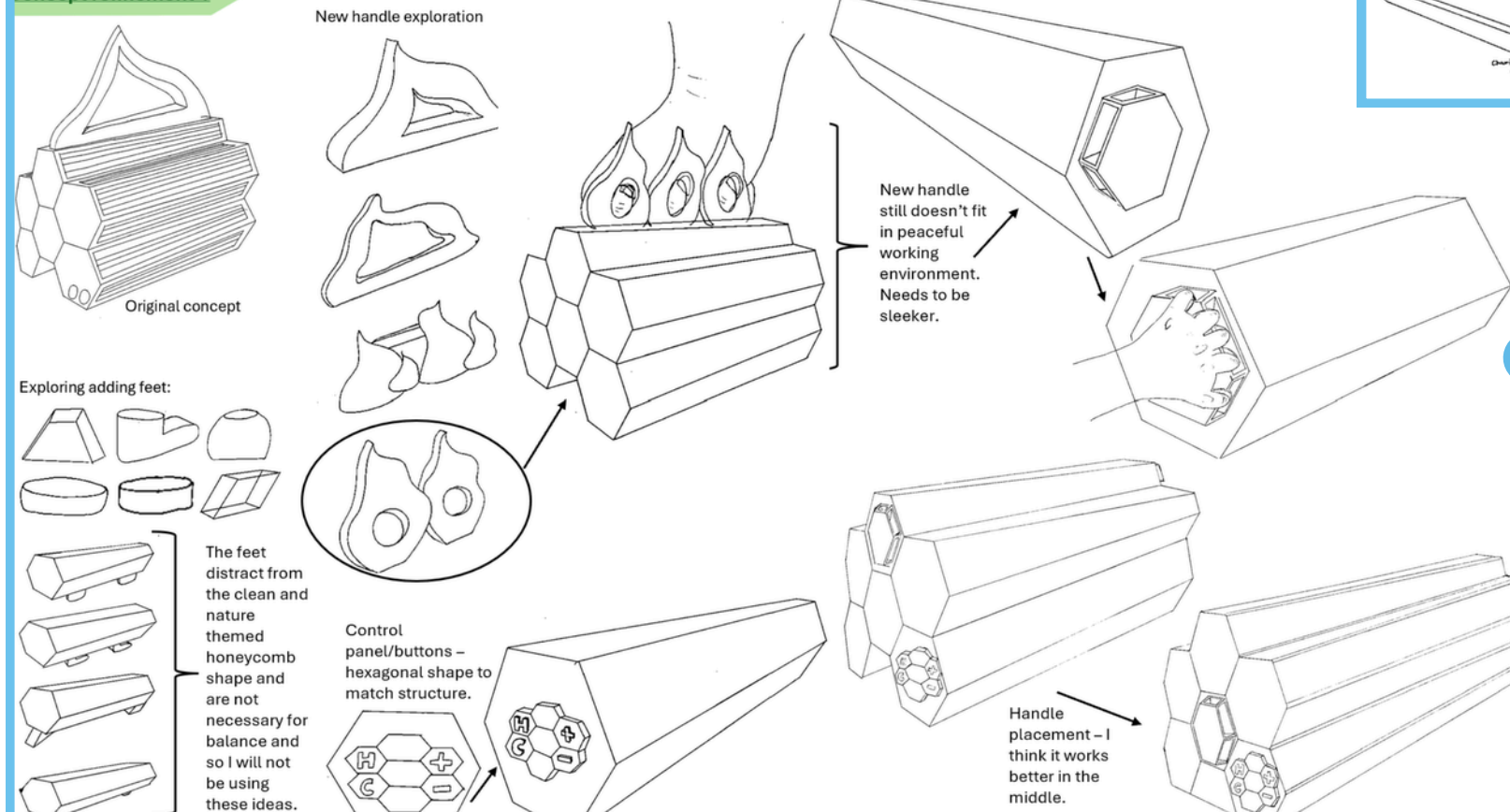


Refinements for machining. Split line down middle with snap fittings.

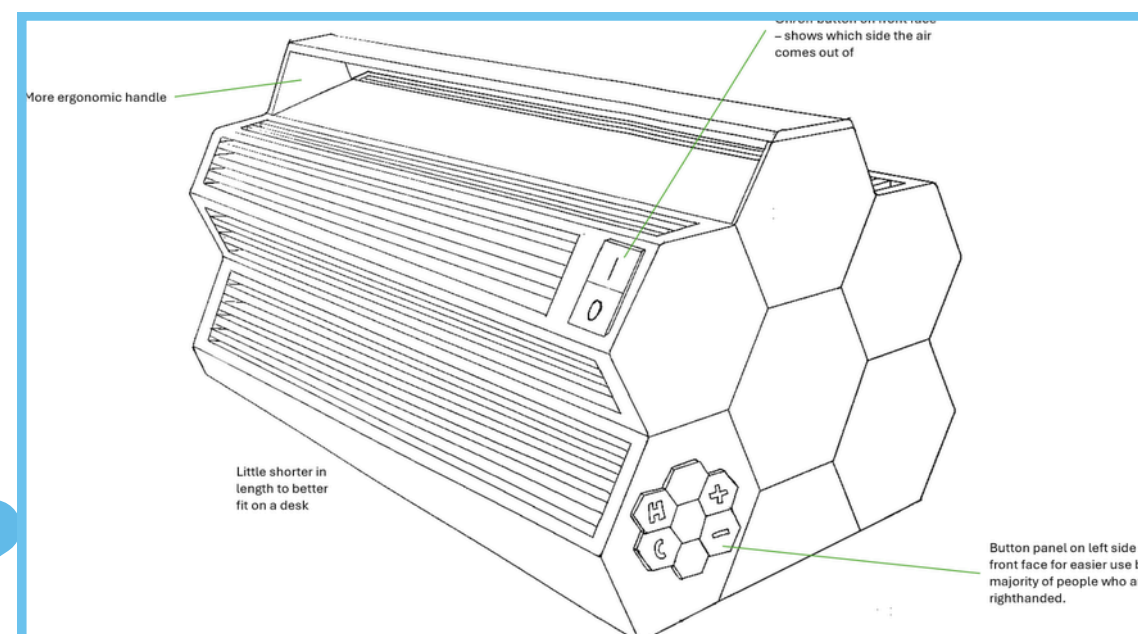


Concept refinement page one

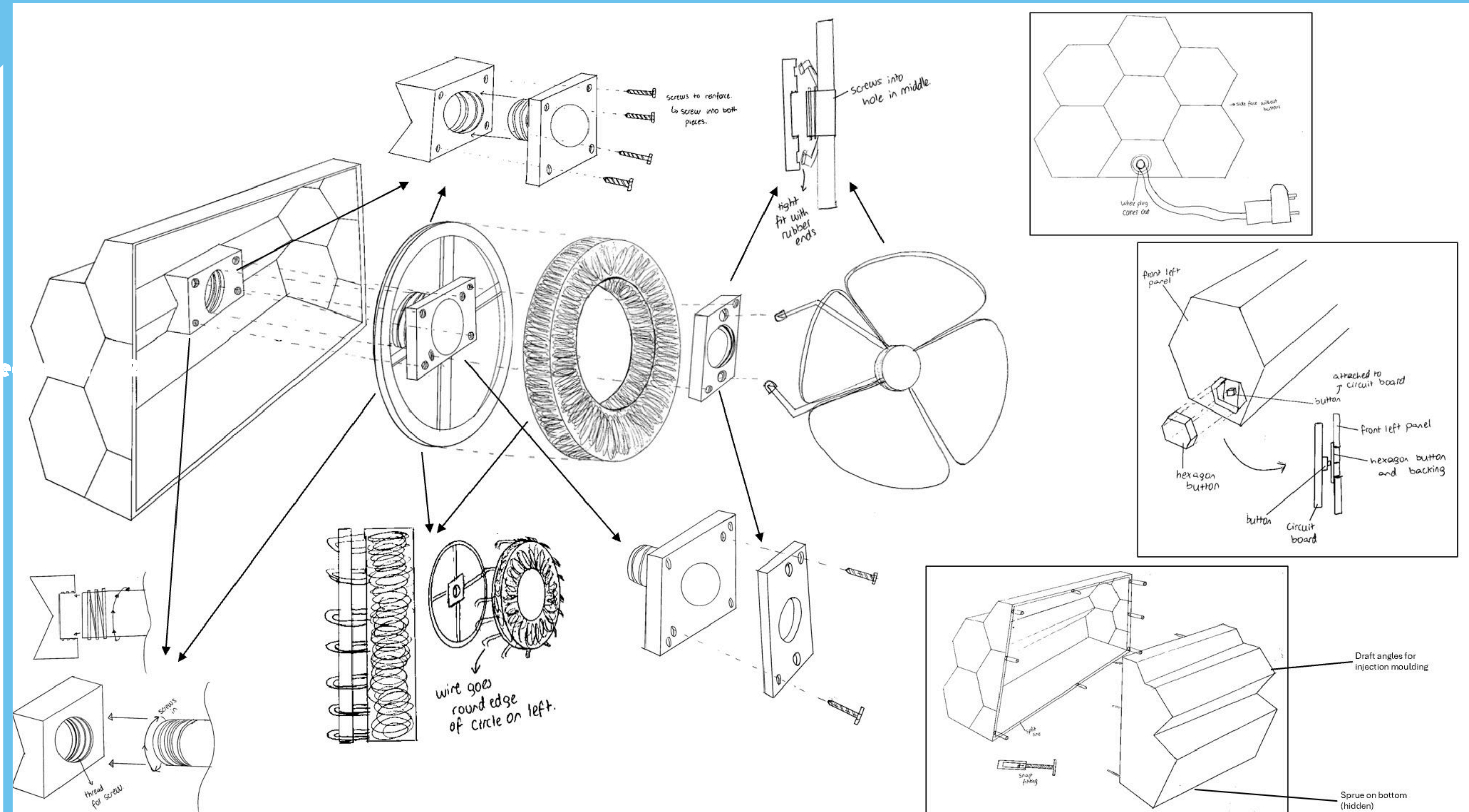
Concept refinement 1



Concept refinement page two



Final refinements

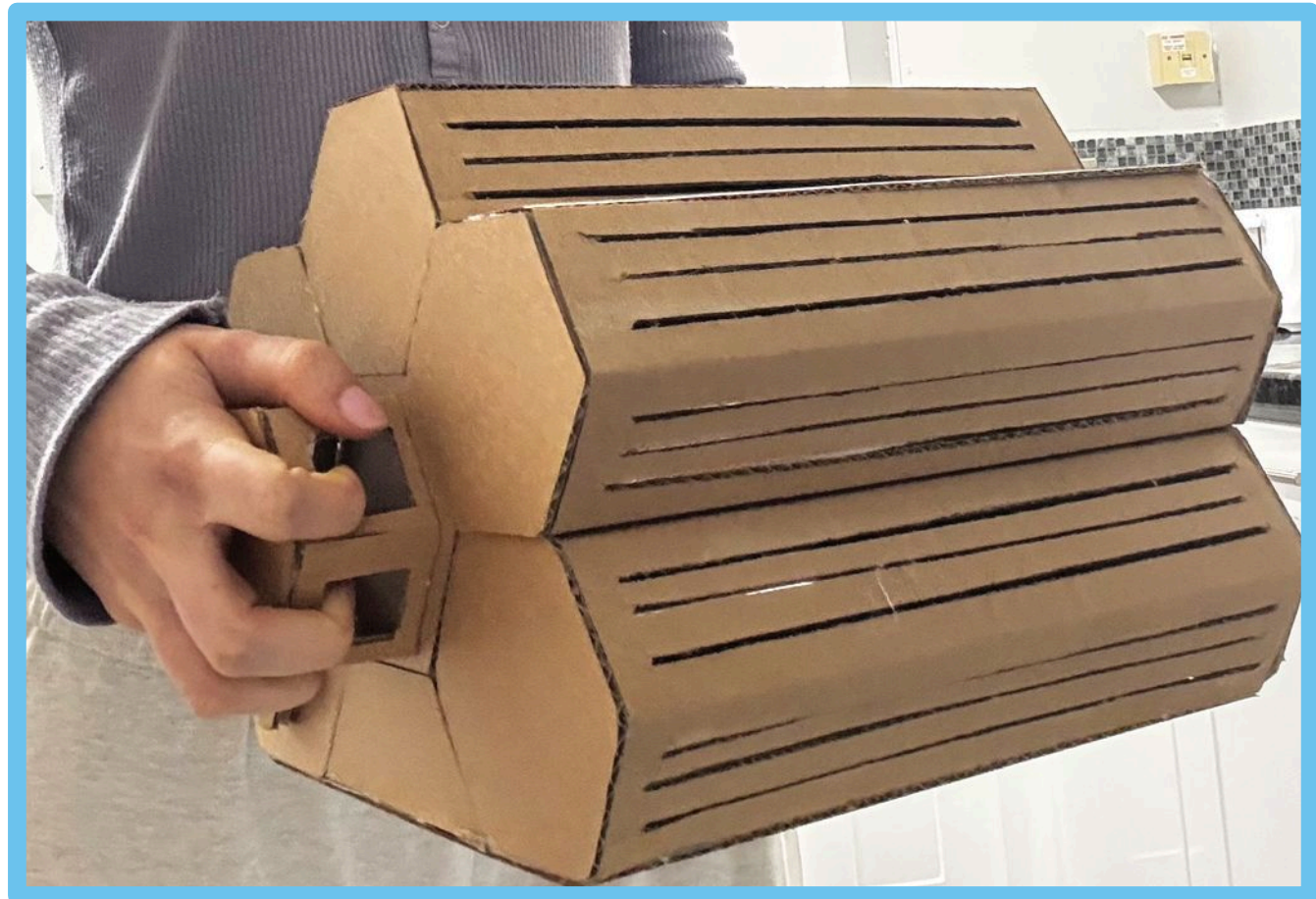


Exploded diagram showing inner workings of the heater and connection method of the parts. Used a heater that I took apart myself to see how the mechanics work.

Cardboard modelling set up: scalpel, ruler, mat



All pieces cut using template before gluing



Final cardboard model of heater.

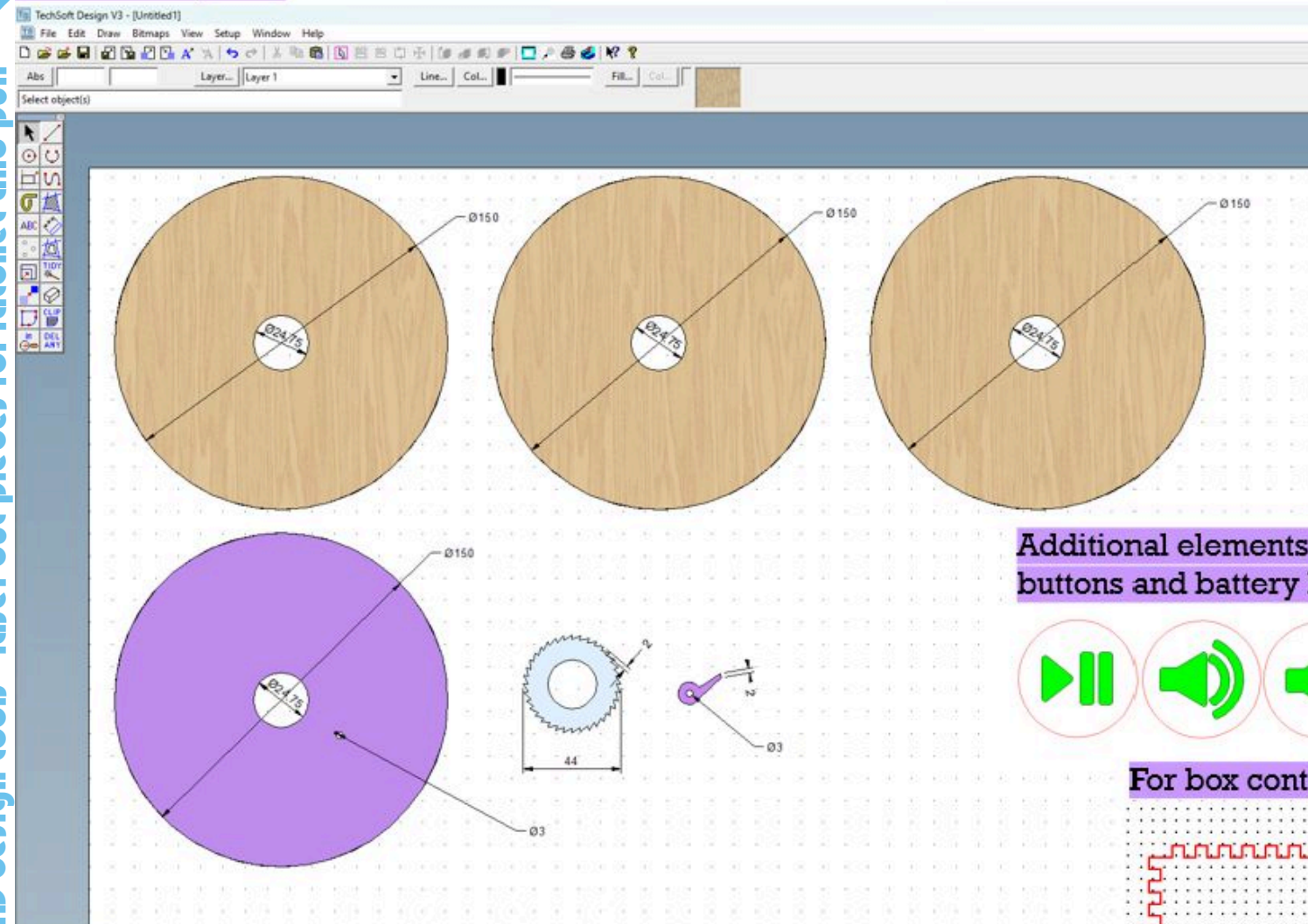
Shows ergonomic usage and detailed buttons. Demonstrated to me aesthetics in shape and size which is hard to see through drawing. I liked the soft edges of the product.



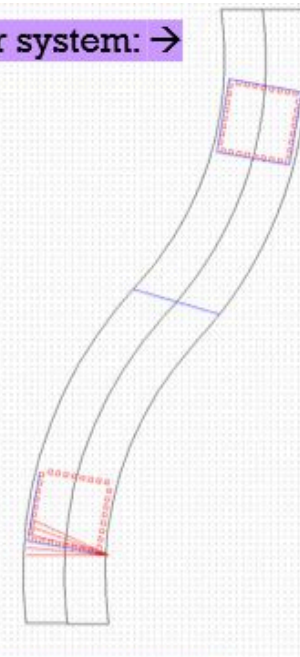


Sensory walkway

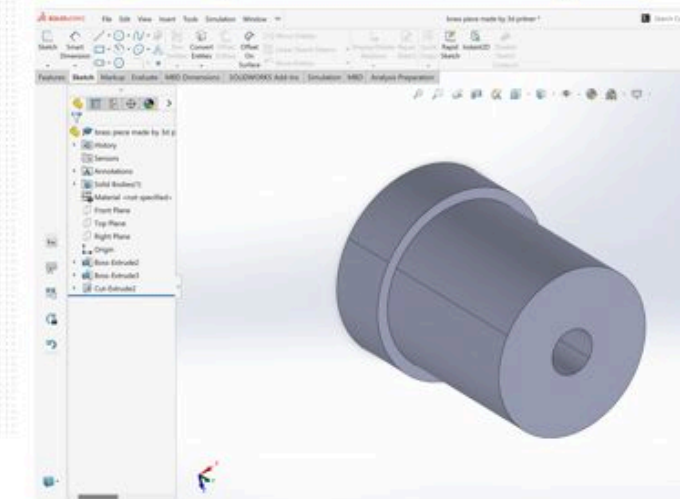
For ratchet system:



For platform holding speaker system: →



For spinning mechanism – to 3d print:

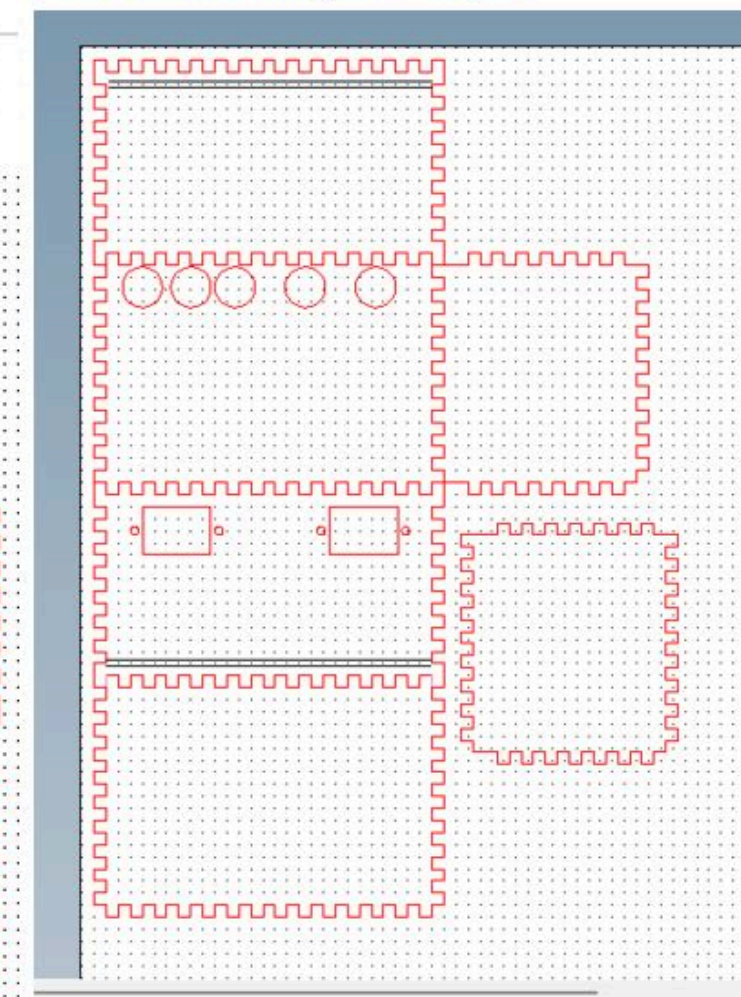


Solidworks component

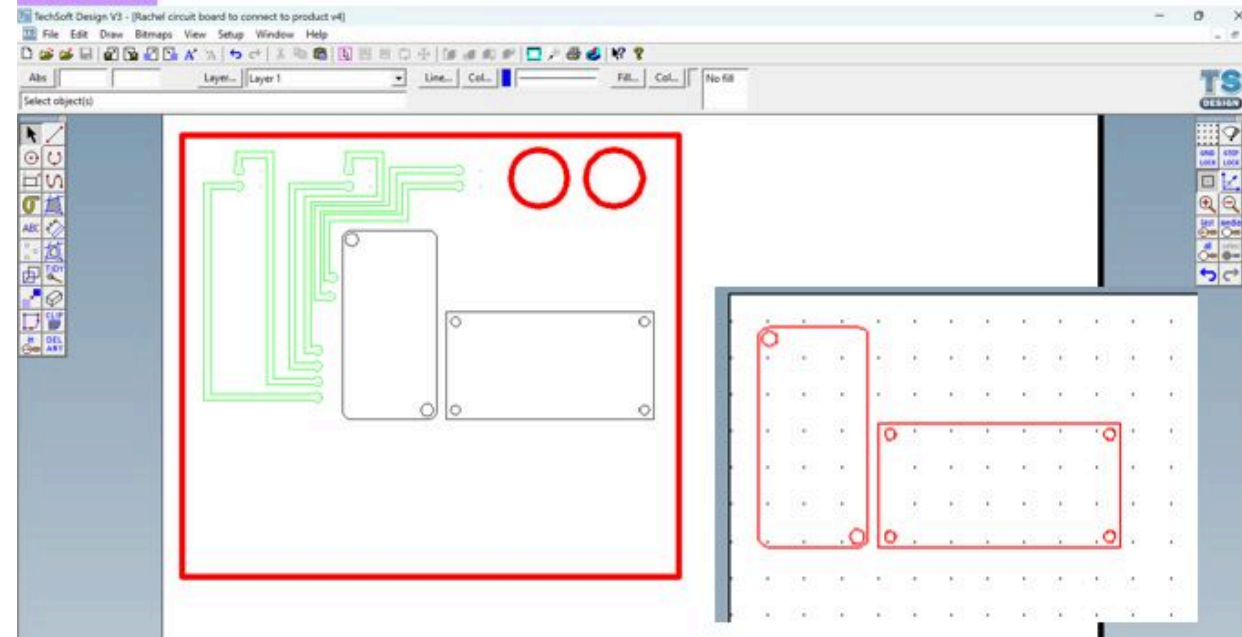
Additional elements of systems box – buttons and battery holder:



For box containing circuitry: :

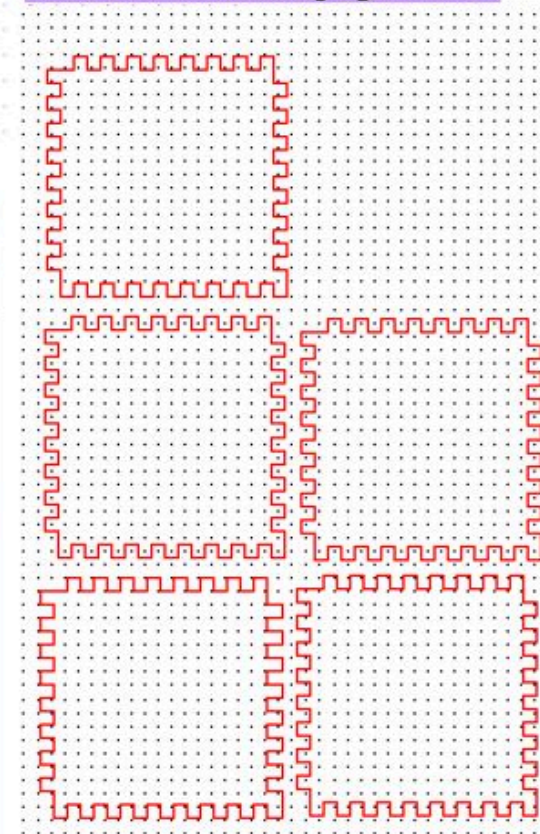


For PCB:



2D solidworks - copper circuit board. Made using CAM.

For box containing speakers:

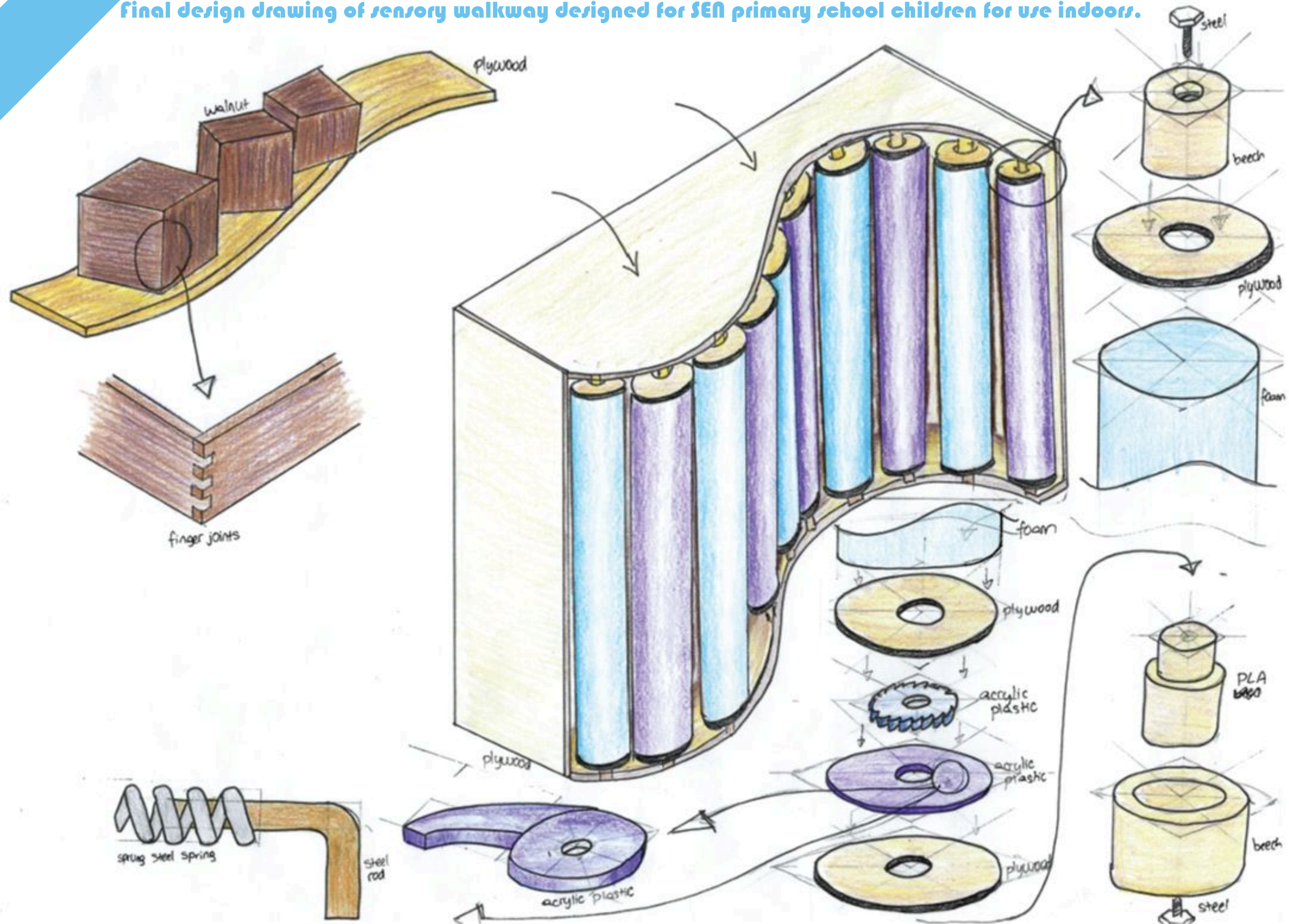


Finger-jointed box on 2D design tools.

Project work: sensory walkway - final design drawing

16

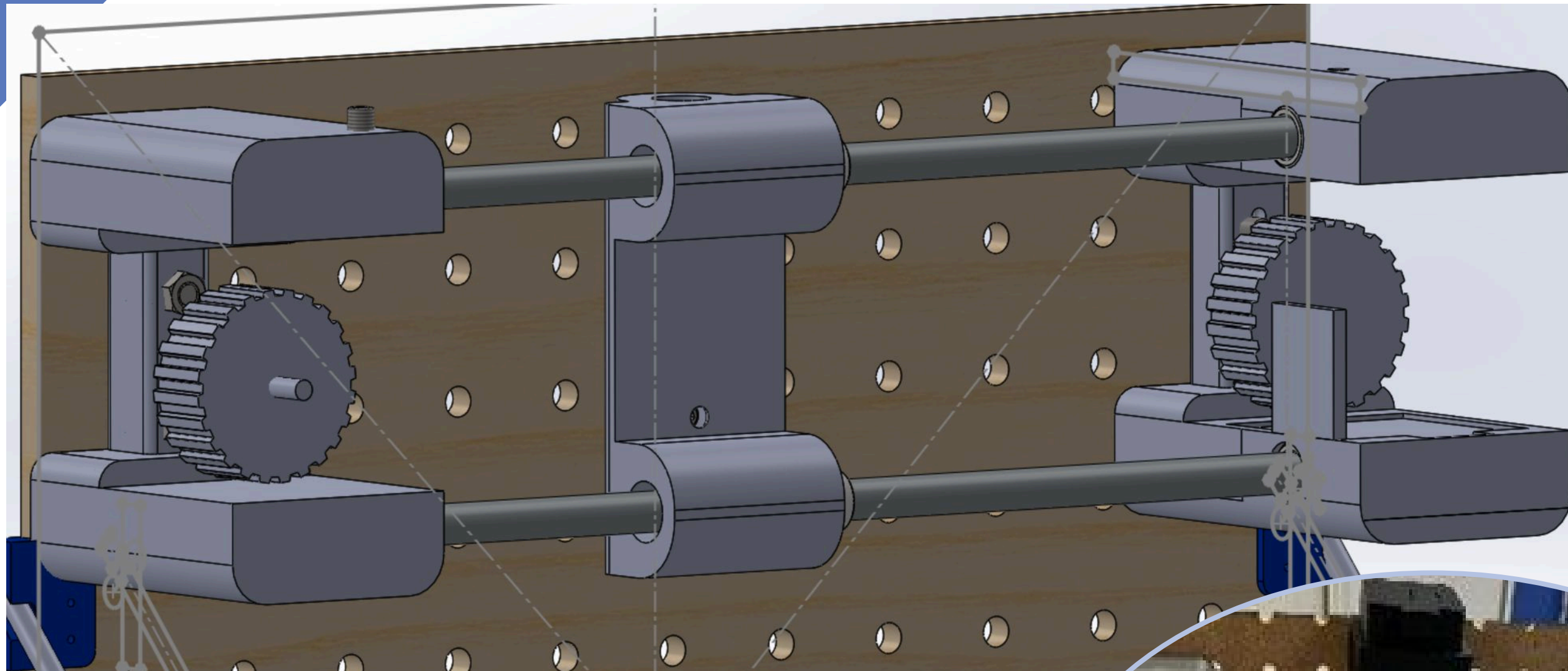
Final design drawing of sensory walkway designed for SEN primary school children for use indoors.





Use of my product by stakeholders. These are children of similar age to those who would use the product. I interviewed them numerous times throughout the design stage, and at the end to evaluate. The product was a success in that they enjoyed interacting with it.

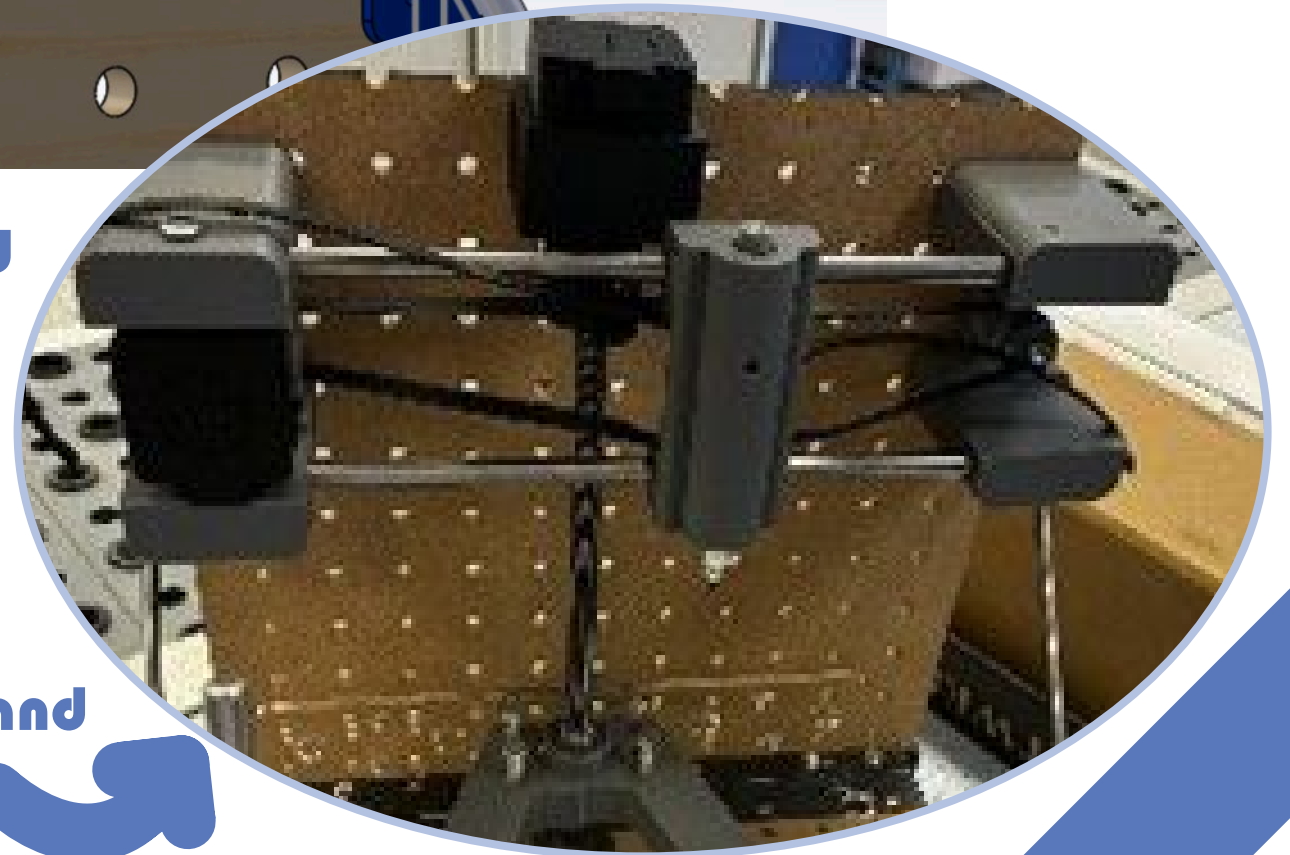
Technical skills

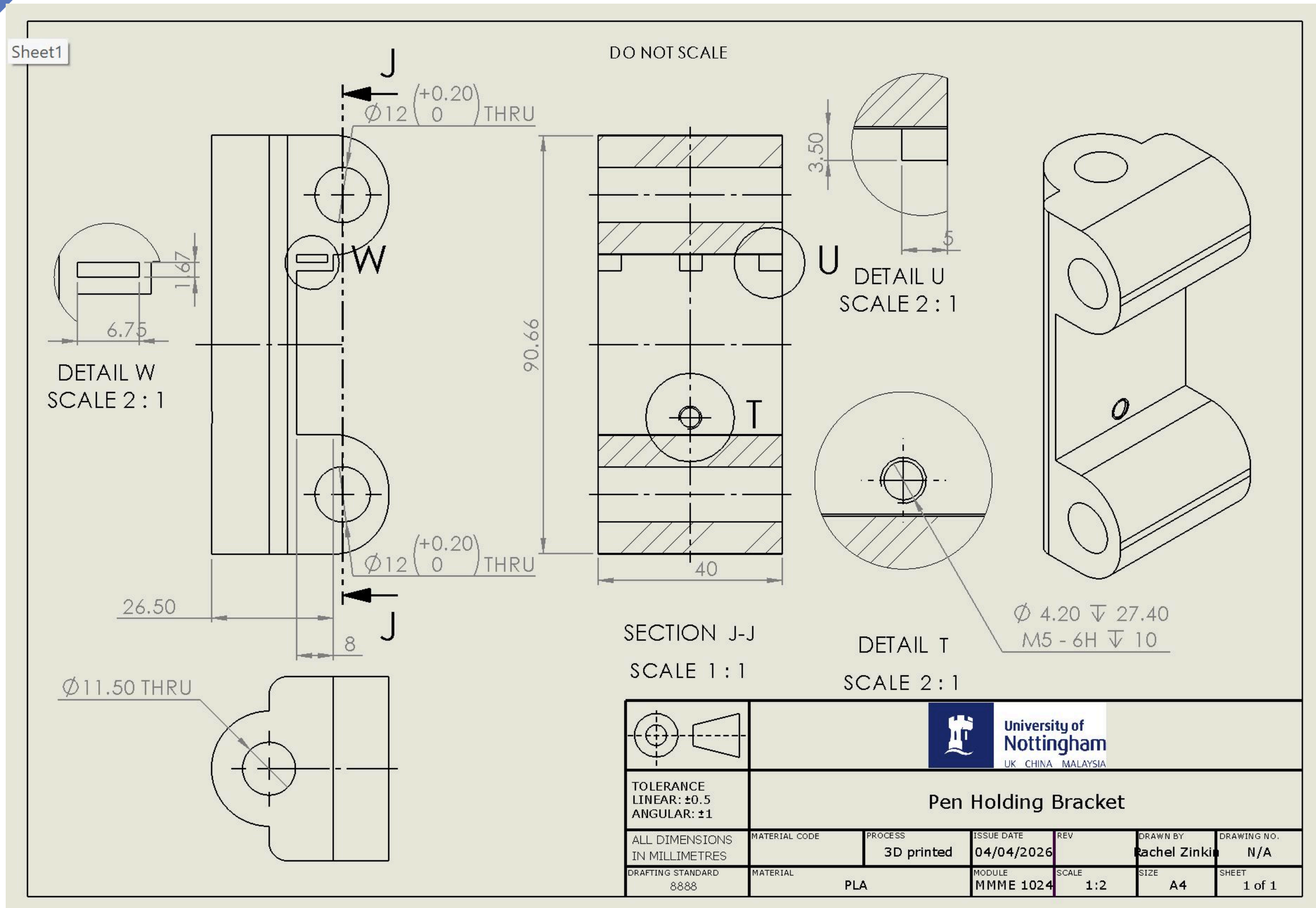


The parts were designed separately and then I used the assembly feature on Solidworks to put it together.

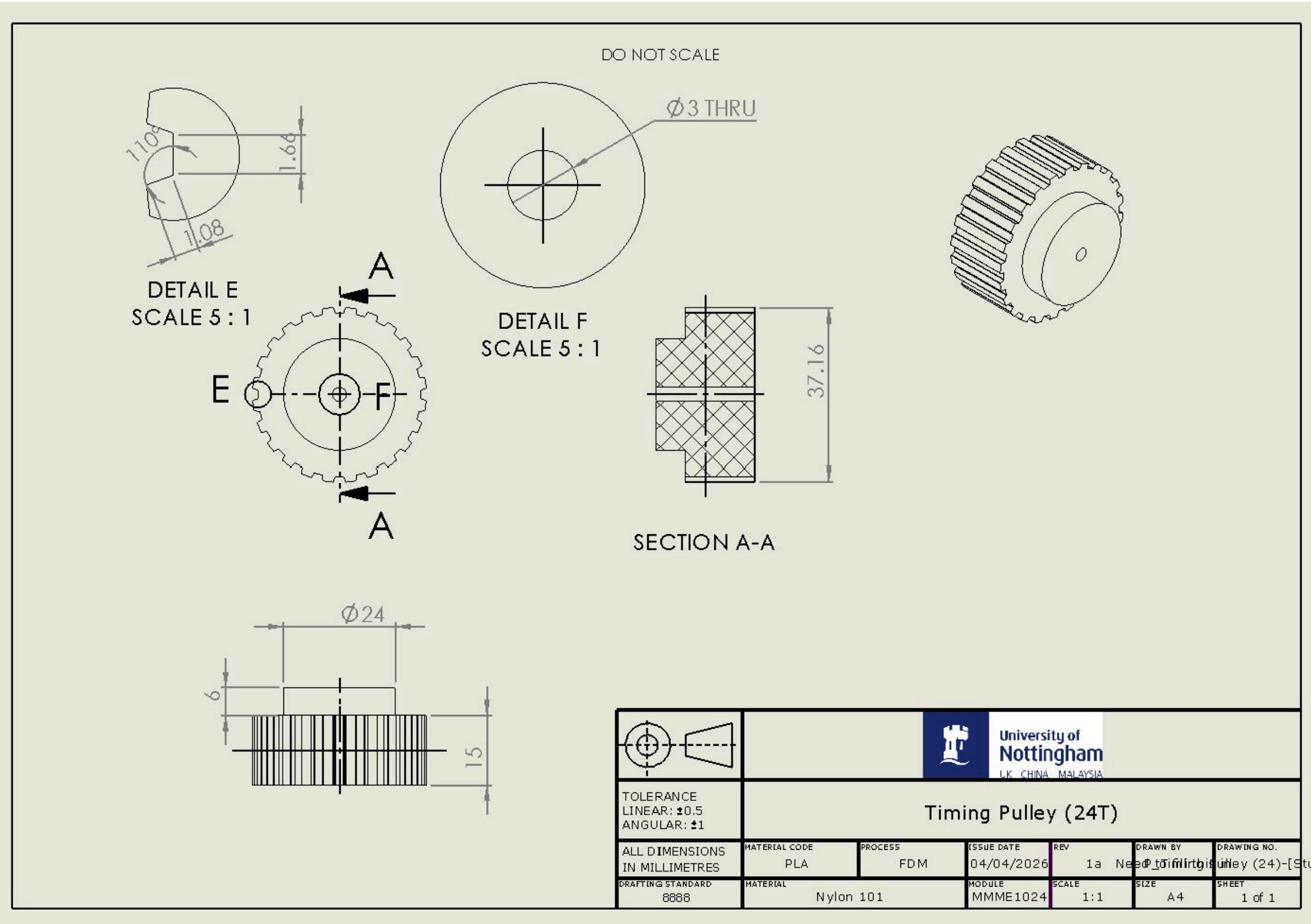
Belt and pulley system to move middle bracket left and right. I designed this using solidworks and some standard components provided to me. The three big brackets were all designed by me and my partner. The middle one holds a pen, and as it slides left and right a line is drawn.

The finished system is shown here complete with motors and belt.



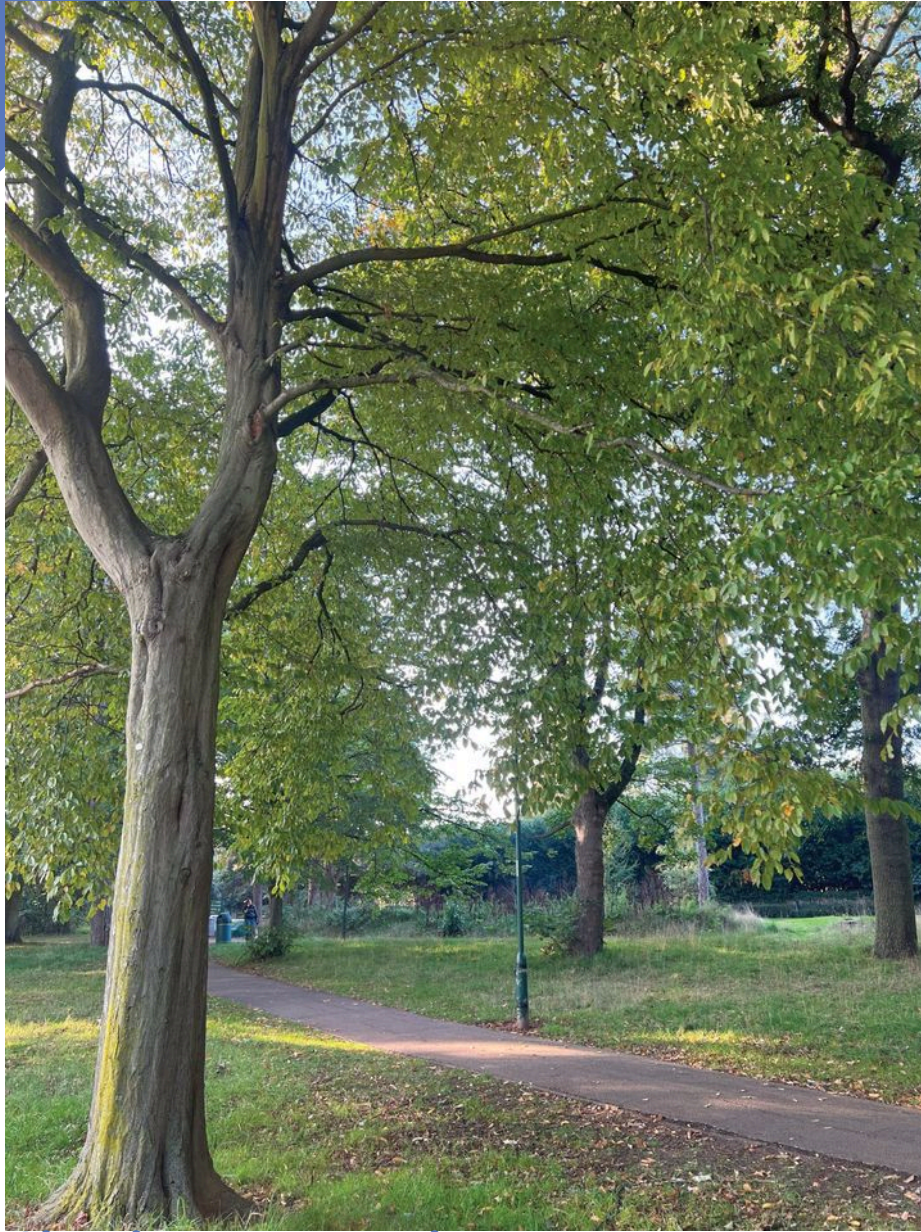


Detailed drawing of 3D-printed component of above system.



Detailed drawing of 3D-printed standard component of above system.

Photography assignment: exploration using the rule of thirds - 3 different themes.



finding letters in nature



Exploration of light



Architecture on campus



Practising cardboard modelling before prototyping my own designs.